CBSE Class 10 Mathematics — Coordinate Geometry — Solutions

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1. **1.** The distance of the point $P(3,4)$ from the origin is:

   Answer: (C)

   Solution:

   1. Use the distance formula from the origin $d = \sqrt{x^2 + y^2}$
   2. Substitute $x = 3$ and $y = 4$
   3. Calculate $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

   Derivation:

   $$d = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

   Coordinate Geometry — Distance Formula

2. **2.** The coordinates of the midpoint of the line segment joining the points $A(2,4)$ and $B(4,6)$ are:

   Answer: (B)

   Solution:

   1. Use the midpoint formula $M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
   2. Substitute the values: $x_1 = 2, x_2 = 4, y_1 = 4, y_2 = 6$
   3. Calculate: $x = \frac{2+4}{2} = 3$ and $y = \frac{4+6}{2} = 5$

   Derivation:

   $$M = (\frac{2+4}{2}, \frac{4+6}{2}) = (3,5)$$

   Coordinate Geometry — Section Formula

3. **3.** The abscissa of the point $(5,8)$ is:

   Answer: (A)

   Solution:

   1. Recall that in a coordinate point (x,y), x is the abscissa and y is the ordinate.
   2. For the point $(5,8)$, the x-coordinate is 5.

   Derivation:

   $$Abscissa = x = 5$$

   Coordinate Geometry — Introduction to Cartesian Plane

4. **4.** If the distance between points $(2,-2)$ and $(-1,x)$ is 5, then the value of x is not asked, but what is the distance formula expression?

   Answer: (D)

   Solution:

1. The distance formula between two points (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
2. Substitute $x_1 = 2, y_1 = -2, x_2 = -1, y_2 = x$

Derivation:

$$d = \sqrt{(-1 - 2)^2 + (x - (-2))^2} = \sqrt{(-3)^2 + (x + 2)^2}$$

Coordinate Geometry — Distance Formula

5. **5.** The point which lies on the y-axis is:

Answer: (C)

Solution:

1. A point on the y-axis has its x-coordinate equal to 0.
2. Check the options for $x = 0$.

Derivation:

$$Pointony-axisisoftheform(0, y).$$

Coordinate Geometry — Introduction to Cartesian Plane

6. **6.** Assertion (A): The point $(0, 5)$ lies on the y-axis. Reason (R): For any point on the y-axis, the x-coordinate is zero.

Answer: (A)

Solution:

1. Check the definition of y-axis points.
2. Verify if (0, 5) satisfies the condition x=0.
3. Conclude that R explains A.

Derivation:

$$Apoint(x, y)liesonthey{-}axisifandonlyifx = 0.Since(0, 5)hasx = 0, itliesonthey{-}axis.$$

Coordinate Geometry — Introduction to Coordinate Geometry

7. **7.** Assertion (A): The distance of the point $P(3, 4)$ from the origin $(0, 0)$ is 5 units. Reason (R): The distance of a point (x, y) from the origin is calculated by $\sqrt{x^2 + y^2}$.

Answer: (A)

Solution:

1. Apply the distance formula from the origin.
2. Substitute x=3 and y=4.
3. Calculate $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

Derivation:

$$d = \sqrt{(3 - 0)^2 + (4 - 0)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

Coordinate Geometry — Distance Formula

8. **8.** Assertion (A): The midpoint of the line segment joining $(2, 4)$ and $(6, 8)$ is $(4, 6)$. Reason (R): The midpoint formula is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.

Answer: (A)

Solution:

1. Identify coordinates $(x_1, y_1) = (2, 4)$ and $(x_2, y_2) = (6, 8)$.
2. Use the midpoint formula.
3. Calculate $(\frac{2+6}{2}, \frac{4+8}{2}) = (4, 6)$.

Derivation:

$$M = (\frac{2+6}{2}, \frac{4+8}{2}) = (\frac{8}{2}, \frac{12}{2}) = (4, 6)$$

Coordinate Geometry — Section Formula/Midpoint

9. **9.** Assertion (A): The points $(1, 1), (2, 2)$ and $(3, 3)$ are collinear. Reason (R): Three points are collinear if the sum of distances between any two pairs is equal to the distance between the third pair.

Answer: (B)

Solution:

1. Check the slopes: slope between (1,1) and (2,2) is (2-1)/(2-1) = 1.
2. Slope between (2,2) and (3,3) is (3-2)/(3-2) = 1.
3. Since slopes are equal, they are collinear.
4. R is a true statement, but typically collinearity is defined by slope or area = 0 in Class 10.

Derivation:

$$Slope_{AB} = 1, Slope_{BC} = 1. Since slopes are equal, points lie on the same line.$$

Coordinate Geometry — Collinearity

10. **10.** Assertion (A): The area of a triangle with vertices $(0, 0), (4, 0)$, and $(0, 3)$ is 6 sq units. Reason (R): Area of a triangle with coordinates $(0, 0), (x_1, y_1), (x_2, y_2)$ is $\frac{1}{2}|x_1y_2 - x_2y_1|$.

Answer: (A)

Solution:

1. Use the simplified area formula for a triangle with one vertex at origin.
2. Substitute $(x_1, y_1) = (4, 0)$ and $(x_2, y_2) = (0, 3)$.
3. Calculate $\frac{1}{2}|(4)(3) - (0)(0)| = \frac{1}{2}(12) = 6$.

Derivation:

$$Area = \frac{1}{2}|4 \times 3 - 0 \times 0| = 6$$

Coordinate Geometry — Area of Triangle

11. 11. Find the ratio in which the line segment joining the points $A(-3, 10)$ and $B(6, -8)$ is divided by the point $P(-1, 6)$.

Answer:

$$2 : 7$$

Solution:

1. Use the section formula: $P(x, y) = \left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$
2. Substitute the coordinates and assume ratio $k : 1$
3. Solve for k using the x-coordinate equation
4. Verify with the y-coordinate

Derivation:

$$\begin{aligned} -1 &= \frac{k(6) + 1(-3)}{k+1} \\ \implies -k - 1 &= 6k - 3 \\ \implies 2 &= 7k \\ \implies k &= \frac{2}{7} \end{aligned}$$

Coordinate Geometry — Section Formula

12. 12. If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p.

Answer:

$$p = 7$$

Solution:

1. Recall that diagonals of a parallelogram bisect each other.
2. Find the midpoint of diagonal AC.
3. Find the midpoint of diagonal BD.
4. Equate the midpoints and solve for p.

Derivation:

$$Midpoint of AC = (\frac{6+9}{2}, \frac{1+4}{2}) = (7.5, 2.5); Midpoint of BD = (\frac{8+p}{2}, \frac{2+3}{2}) = (\frac{8+p}{2}, 2.5); \frac{8+}{2}$$
$$\implies 8 +$$
$$\implies$$

Coordinate Geometry — Midpoint Formula

13. 13. Find the distance between the points $P(a + b, b - a)$ and $Q(a - b, a + b)$.

Answer:

$$2\sqrt{a^2 + b^2}$$

Solution:

1. Apply the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

2. Substitute the given coordinates.
3. Simplify the algebraic terms inside the square root.
4. Calculate the square root.

Derivation:

$$d = \sqrt{((a-b)-(a+b))^2 + ((a+b)-(b-a))^2} = \sqrt{(-2b)^2 + (2a)^2} = \sqrt{4b^2 + 4a^2} = 2\sqrt{a^2+b^2}$$

Coordinate Geometry — Distance Formula

14. **14.** Find the value of y for which the distance between the points $P(2,-3)$ and $Q(10,y)$ is 10 units.

Answer:

$$y = 3ory = -9$$

Solution:

1. Use the distance formula $PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
2. Substitute the known values into the equation.
3. Solve the resulting quadratic equation for y.

Derivation:

$$\begin{aligned} 10^2 &= (10-2)^2 + (y-(-3))^2 \\ \implies 100 &= 64 + (y+3)^2 \\ \implies 36 &= (y+3)^2 \\ \implies y+3 &= \pm 6 \\ \implies y &= 3, -9 \end{aligned}$$

Coordinate Geometry — Distance Formula

15. **15.** Find the coordinates of a point A, where AB is the diameter of a circle whose centre is $(2,-3)$ and B is $(1,4)$.

Answer:

$$A(3,-10)$$

Solution:

1. Let A be (x,y). The centre $(2,-3)$ is the midpoint of diameter AB.
2. Use the midpoint formula: $\frac{x+1}{2} = 2$ and $\frac{y+4}{2} = -3$.
3. Solve for x and y.

Derivation:

$$\begin{aligned} x+1 &= 4 \\ \implies x &= 3; y+4 = -6 \\ \implies y &= -10. \text{Point A is } (3,-10) \end{aligned}$$

Coordinate Geometry — Midpoint Formula

16. **16.** Find the ratio in which the point $P(x, 2)$ divides the line segment joining the points $A(12, 5)$ and $B(4, -3)$. Also, find the value of x.

Answer:

$$Ratiois 3 : 5, x = 9$$

Solution:

1. Use the section formula for the y-coordinate to find the ratio k:1.
2. Set the y-coordinate formula equal to 2.
3. Solve for k.
4. Use the value of k to find the x-coordinate using the section formula.
5. State the final ratio and the value of x.

Derivation:

$$Lettheratiobek : 1. They - coordinateisgivenbyy = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}. Substitutingvalues : 2 = \frac{k(-3) +}{k +}$$
$$\implies 2k + 2 = -3$$
$$\implies 5$$
$$\implies k = 3/5. Thus, theratiois 3 : 5. For x : x = \frac{3(4) + 5(12)}{3 + 5} = \frac{12 + 60}{8} = \frac{72}{8}$$

Coordinate Geometry — Section Formula

17. **17.** If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p.

Answer:

$$p = 7$$

Solution:

1. Recall the property that diagonals of a parallelogram bisect each other.
2. Find the midpoint of diagonal AC.
3. Find the midpoint of diagonal BD in terms of p.
4. Equate the midpoints and solve for p.

Derivation:

$$Midpoint of AC = (\frac{6 + 9}{2}, \frac{1 + 4}{2}) = (7.5, 2.5). Midpoint of BD = (\frac{8 + p}{2}, \frac{2 + 3}{2}) = (\frac{8 + p}{2}, 2.5). Equa$$

Coordinate Geometry — Midpoint Formula

18. **18.** Find the area of a quadrilateral whose vertices, taken in order, are $(-4, -2)$, $(-3, -5)$, $(3, -2)$, and $(2, 3)$.

Answer:

$$28 sq.units$$

Solution:

1. Divide the quadrilateral into two triangles by drawing a diagonal.
2. Use the area of triangle formula: $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
3. Calculate the area of triangle 1.
4. Calculate the area of triangle 2.
5. Add the two areas to get the total area.

Derivation:

$$Area of \triangle ABC with vertices (-4,-2),(-3,-5),(3,-2): \frac{1}{2}|-4(-5-(-2))+(-3)(-2-(-2))+3(-2-$$

Coordinate Geometry — Area of Triangle

19. **19.** Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.

Answer:

$$(-1, 3.5), (0, 5), (1, 6.5)$$

Solution:

1. Identify the midpoints needed to divide the segment into 4 parts.
2. Find the midpoint M of AB.
3. Find the midpoint of AM to get the first point.
4. Find the midpoint of MB to get the third point.
5. List all three points.

Derivation:

$$Midpoint M of AB = (\frac{-2+2}{2}, \frac{2+8}{2}) = (0, 5). Midpoint P of AM = (\frac{-2+0}{2}, \frac{2+5}{2}) = (-1, 3.5). Mi$$

Coordinate Geometry — Section Formula Application